

Interview Questions

Questions I Have Been Asked

- If you were working on a project to develop a new type of medical grade adhesive, what properties would you investigate and how would you investigate them?
- Tell me about a time you spearheaded a process improvement?
- How have your communication skills improved in terms of your management roles as a supervisor?

Practice Questions

Engineering Questions

What sort of design engineering experience do you have?

- Main focus of my last 2 years of undergraduate study was medical device design
 - Capstone project
 - Design engineering tools:
 - Design process control map
 - User needs
 - Device inputs
 - Device outputs
 - Verification (outputs to inputs)
 - Validation (user needs to outputs)
 - Design History File compliant with 21 CFR 820 and ISO 13485
 - Knowledge of standards:
 - 21 CFR 820
 - 21 CFR 1271
 - ISO 13485
 - American Association of Tissue Banks (now Association for Advancement of Tissue and Biologics) standards
 - Prototype device won school award for computerized systems

Describe a technical problem that you solved

- In process test for residual sodium hypochlorite began failing consistently, put a halt in our production
- Used batch records to collect information and determine what each failed test had in common
- Used test result documentation to establish a timeline and chronological trend for the failed test results
- Observed production methods and in process testing methods and materials, discovered that the incorrect reagent had been used for conducting in process testing
- Established root cause – material spec sheet for new test kits did not include inspection of the included reagent part number, part number was only present for the test kit itself
- Generated NCR, CAPA, and new material spec sheet, and spearheaded training procedures and implementation and review of the CAPA

How have you used statistical analysis in the past?

- [ANOVA Test: An In-Depth Guide with Examples | DataCamp](#)
- capstone project; we performed ANOVA calculations for our quantitative tests to optimize our prototype
- Used ANOVA to compare results of different computational algorithms for determining position from acceleration data
- Used paired t-test for analyzing changes to one algorithm step by step
- Used two way ANOVA to compare several different base algorithms each with two distinct phases of computation
- At Stability I spearheaded a data collection program to create performance metrics for my team and used paired t-tests to establish evidence of performance improvement. I recommended a performance bonus incentive based on this data.
 - Used this information to make data-driven decisions regarding delegation of responsibilities
 - There was a lack of performance metrics when I joined the team

Can you discuss your experience with process control and improvement?

- I have worked on several different document change requests
- I spearheaded a project to automate a lot of our labeling information to remove user error that was created by technicians.
 - This involved creating a process map for the new algorithms

- Writing a validation protocol report for the new process including statistical proof that the protocol created accurate labels
- I have written the validation protocol for a new manufacturing process that changed the disinfection phase of our birth tissue allografts. This new process reduced the amount of time it took for the wash phase, the validation involved bioburden testing which we contracted outside of our company, and it was my responsibility to write the validation protocol and determine that the new process met our user needs/design specifications for disinfection.

What methods would you use to ensure that a product meets its design specifications?

- I would make sure that performance criteria are strong and consistent
- I would use regular manufacturing observations to ensure that best practices are being followed
- I would establish in process testing criteria to determine that the product meets the design specifications at different phases in processing
- I would utilize statistically validated sampling procedures and rigorous product inspections to ensure the highest quality is achieved.

Can you describe a situation where you had to balance the needs of the customer with the constraints of the company?

- During a significant growth phase last year our company customer demand outpaced production capacity and we were unable to meet our demand. During this period extensive overtime was necessary, it was an all hands on deck situation, and many people were upset. I counterbalanced this decline on morale by advocating for a performance based bonus system for my team members.

Could you outline your experience with quality systems?

- I was directly responsible for ensuring compliance with all standards set forth by the American Association of Tissue Banks and by the FDA 21 CFR 1271 regulations
- I have worked extensively with document control, labeling control, and process controls within these frameworks
- I have written document change requests, non conformance reports, validation protocol reports, IQ OQ PQ reports, and I have spearheaded

process optimizations involving validation of process changes within the quality management system.

- I also worked on applying six sigma lean principles to make continuous improvements to the quality control processes with changes such as new pull systems, redesigned work areas, and more efficient communication strategies.

How would you handle a situation where a product failed to meet its performance specifications?

- Quarantine the product and observe processes and well as the quality control systems in place
- I would analyze testing data from previous lots to determine if there was any detectable trend and what the context was in terms of how long that trend was going on and how significantly the results changed
- I would open the design history file and determine if the failure mode was consistent with anything contained in the risk probability matrix or failure mode and effects analysis
- If none of this information led to a conclusive fix then I would implement more in process evaluations earlier in the process to determine where the failure mode was being introduced and I would document in process evaluations heavily
- If no solution was arrived at, I would reassess the risk priority number and design history file and see if there was a different way to achieve the required performance specification.

What are your strategies for managing and prioritizing multiple projects simultaneously?

- I like to use priority matrices and inter-project gantt charts to plan my management of multiple projects
- I use work breakdown structures to define tasks into subtasks until I feel as though I can predict the amount of time a task will take, then I place those subtasks in chronological order within a gantt chart
- I use a priority matrix to determine in which order I should complete tasks that can be done simultaneously
- In my current management context I use the priority matrix to determine what I can delegate and who I can delegate it to.

Please describe your experience with root cause analysis and corrective action processes.

- In my current role I have carried out root cause analysis using data trend and batch records to determine when a problem began happening and what do the problem results all have in common. One example is my experience with the determination that my team was using the incorrect reagent for an in process laboratory test. I used batch records to determine what they all had in common, then I used our QC internal data collection to determine when the incorrect reagent was introduced, which is how I came to the conclusion that the incorrect reagent was introduced at the same time that we received a brand new test kit.

How have you ensured that a product adheres to all relevant safety and regulatory standards in a previous role?

- I maintain an active continuing education program for me and my team members to stay up to date on relevant safety and regulatory requirements. This includes reviewing our applicable standards yearly and including the discussion of any updated standards in our team meetings.

Can you provide an example of a time when you had to make a difficult decision that had a significant impact on product sustainability?

- There was an occasion where the quality assurance team came to us because they were missing testing records for some of our lots, and they requested that we generate new records for them. My team was unable to verify that the testing had ever been conducted for those lots, and ultimately I had to tell the quality assurance team that no record existed and the lots would need to be scrapped and an NCR generated. This was during a very busy growth phase where we were already not meeting customer demand and this decision put all of the departments in a difficult position in terms of conducting root cause analysis to see who was at fault. Ultimately the process through which manufacturing submits in process testing samples was changed.

How would you deal with a situation where a product design needed to be changed due to unforeseen issues?

- I would generate a new design specification based on the issues that were discovered. I would start by creating design inputs based on the new issue

and discuss with other departments or with customers to determine if the user needs or intended use of the device needed to be adjusted. I would design a testing protocol to assess whether the changes made to the device satisfy the new design inputs.

Can you discuss a time when you used data analysis to improve a product's performance or reliability?

- Product will meet all specifications pre-sterilization but comes back and does not pass the post-sterilization evaluation requirements
- We have analyzed the donor records and established a solid trend relating donor characteristics to the lots that fail post-ster evaluation
- We are currently still developing new processing procedures to account for variation in donor characteristics based on the trend we have established.

Resume Questions

Tell me about your work at Stability

- I have experience working in both manufacturing and quality control at stability.
- I have gained significant knowledge of the AATB standards and FDA regulations from 21 CFR 1271
- I am knowledgeable regarding current good manufacturing practices (cGMP), aseptic technique, and cleanroom procedures.
- I spearheaded a lot of document change requests, non conformance reports, corrective and preventive action reports, calibrations and IQOQPQ reports, and validation protocol reports.
- I have developed a lot of knowledge regarding processing of birth tissue and bone tissue based biomaterials

Tell me about your laboratory work

- Experience in two different biomaterials labs
- Experience running UV/Vis spectroscopy, NMR spectroscopy, polymer synthesis, cell culturing, and mechanical analysis of different polymers.

Basic/General Questions

Tell me about yourself

- As you can see from my resume I got my bachelors degree in biomedical engineering at Texas A&M

- I am very proud of that education and I am proud to be an Aggie
- I really enjoy working hands on with cutting edge devices and laboratory equipment, and I enjoy learning new things, particularly technical knowledge and skills related to device engineering
- I have a lot of experience working in environments that require extreme attention to detail and good procedural thinking skills
- Outside of work my biggest hobbies are weightlifting and reading
- I also really enjoy travelling and seeing new places, both natural and manmade

What are your strengths?

- I am very good with computers and learning different types of software and I have basic programming skills.
- I am very analytical and have excellent logical and procedural thinking skills.
- I have a lot of basic technical knowledge on many topics from my education as an engineer.
- My analytical skills and background knowledge make me very adept at learning and I think this is my greatest asset; I am able to acquire new technical knowledge and skills very quickly.
- I am naturally inquisitive which is part of why I am able to pick up new skills very quickly, I dive into them on my own time as well as at work.
- I am highly organized; I have good record-keeping and spatial organization skills both at work and at home.

What are your weaknesses?

- One of my biggest weaknesses is my memory; Things like dates and acronyms do not stick in my mind without regular practice, but I compensate for that by staying very well organized. I take notes and set reminders for myself, and I keep all my documents both work and personal very well organized in specific folders and binders with labels on them. I compensate by using tools to extend my memory outside of just my brain.
- Another weakness that comes to mind is speaking anxiety. I do get anxious when making a presentation to a large group of people, especially if I think of those people as leaders or figures of authority. However I really don't let that hold me back. I have had a lot of practice throughout my education and professional career in presenting and I have developed some mental strategies and breathing exercises to overcome the anxiety. I also have

confidence in myself and my technical knowledge so I do not let that anxiety prevent me from sharing my opinions or ideas.

What is your ideal company?

- My ideal company would be one that always gives me the opportunity to learn and to expand upon my knowledge and skillsets.
- I would like a company that supports continued education and creates pathways for growth and improvement
- Related to that I would like to work for a company that is always trying to expand itself and achieve continuous improvement

Why should we hire you?

- You should hire me because I have a passion for learning and an entrepreneurial mindset. I am a self-starter and I commit myself entirely to doing the best job I can no matter what I am doing.
- You should hire me because I am always striving to learn more about my work and to improve my work.
- You should hire me because I combine creative thinking, analytical thinking, and technical knowledge to come up with new ideas and questions to ask to improve the work that I am doing whatever it may be.

Where would you like to be in your career five years from now?

- I would really like to be involved in team management. I want to learn as much as I can about the job I am doing and through that develop the ability to lead others towards success in that area. I think of myself as a great team member and a natural leader
- I would like to develop as many of my own ideas as possible. I want to be working on many different projects, whether that be multiple different improvements to the same product or improvements to multiple different products.

Questions for Me to Ask the Interviewer

- What will be the next step to move forward with this? I am very interested in the role.
- What is the pay range for this position? What makes the difference between a (low end) employee and a (high end) employee?